

phys. stat. sol. (b) 44, 253 (1971)

Subject classification: 11; 21; 21.1; 22.1

*Sektion Physik der Humboldt-Universität zu Berlin,
Bereich Atomstoßprozesse der Festkörperphysik*

Plural and Multiple Scattering of Low-Energy Heavy Particles in Solids

By

L. MEYER¹⁾

Small-angle multiple scattering of low-energy heavy particles (Born parameter $\alpha > 1$) in solid layers is calculated, using a scattering cross section derived from classical mechanics. The angular distribution of the scattered particles is given as a sum of two functions, which are tabulated. The calculated distributions coincide with those of the well-known theory of Molière for large thicknesses of material. For small thicknesses there are profound deviations.

Ausgehend von einem nach der klassischen Mechanik ermittelten Streuquerschnitt wird die Vielfachstreuung niederenergetischer schwerer Teilchen (Born-Parameter $\alpha > 1$) in Festkörperschichten berechnet. Die Rechnung liefert die Winkelverteilung der gestreuten Teilchen als Summe zweier Funktionen, die in Tabellenform angegeben werden. Die berechneten Verteilungen gehen bei großen Schichtdicken in diejenigen der Theorie von Molière über. Bei kleinen Schichtdicken ergeben sich wesentliche Abweichungen von dieser Theorie.

1. Introduction

The interaction between energetic particles and a solid body consists of (i) more or less elastic encounters with the atomic constituents and (ii) of inelastic processes resulting in electronic excitation. The first part of this interaction plays the dominant role in such processes as the scattering of the incident particles, the production of defects in the solid body, cathode sputtering, and the last phase of slowing down of the incident particles. Among these processes, the scattering of the primary particles gives the most direct information about this type of interaction. At low energies this scattering will be predominantly plural and multiple scattering.

Multiple scattering of high-energy particles has been carefully studied in many investigations. Scott [1] gives a detailed review of this work. The theoretical investigations of this process are, however, generally based on differential scattering cross sections which have been derived assuming a small value of the Born parameter, $\alpha = Z_1 Z_2 / (137 \beta)$, i.e. for high-energy particles. Z_1 and Z_2 are the atomic numbers of the incident particles and of the scattering atoms respectively, β is the ratio of the velocity v of the incident particles to the velocity of light. In the present paper the opposite case, $\alpha > 1$, will be considered which means for the energy E of the incident particles

$$E < A_1 Z_1^2 Z_2^2 \times 25 \text{ [keV]}. \quad (1)$$

A_1 is the mass number of the incident particles. The scattering cross section, appropriate to this case, will be discussed in Section 2. Special considerations,

¹⁾ Present address: Deutsche Akademie der Wissenschaften zu Berlin, Zentralinstitut für Optik und Spektroskopie, 1199 Berlin, Rudower Chaussee 6.

reflecting the properties of a solid body as the target material, will be given in Sections 2 and 3.

2. The Scattering Cross Section

In the case $\alpha > 1$ the scattering cross section can be computed by classical mechanics. Computations of this kind have been carried out by several authors, and the results have been compiled in rather extensive tables (see, e.g., [2]). The following calculations are based on the potential

$$V(r) = \frac{Z_1 Z_2 e^2}{r} \varphi\left(\frac{r}{a}\right). \tag{2}$$

The screening function $\varphi(r/a)$ is assumed to be the Thomas-Fermi function. The screening parameter a characterizes the screening of the nuclear charges $Z e$ by the electronic shells and may be computed approximately from the relation

$$a = \frac{0.885 a_0}{(Z_1^{2/3} + Z_2^{2/3})^{1/2}}; \quad a_0 = 0.529 \times 10^{-8} \text{ cm}. \tag{3}$$

From the potential (2) one obtains a scattering cross section $d\sigma/d\theta$ which, besides its dependence on the scattering angle θ , is a function of the reduced energy $\varepsilon = a/b$, where $b = 2 Z_1 Z_2 e^2 (m_1 + m_2)/(m_1 m_2 v^2)$ and m_1 and m_2 are the masses of the incident and target particles.

Lindhard, Nielsen, and Scharff [3] have shown that this dependence of the cross section on two variables (θ and ε) can be reduced in rather good approximation to a dependence on only one quantity $\eta = \varepsilon \sin(\theta/2)$. They obtain the differential scattering cross section in the form

$$\frac{d\sigma}{d\eta} = \pi a^2 \frac{f(\eta)}{\eta^2}. \tag{4}$$

The function $f(\eta)$ is given graphically and by a table. For large values of η this function approaches the function $f_R(\eta) = (2\eta)^{-1}$, which corresponds to the Rutherford cross section. This is shown in Fig. 1. In addition Fig. 1 shows several functions $f_\varepsilon(\eta) = (\eta^2/\pi a^2) (d\sigma/d\eta)$, which have been derived from the exactly computed scattering cross sections $d\sigma/d\eta$ for several energies ε . For

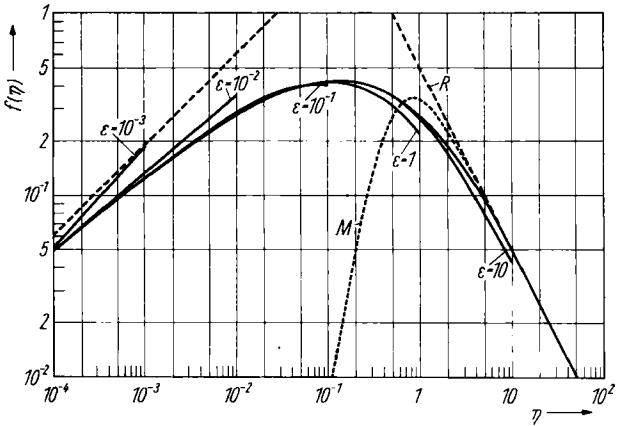


Fig. 1. Scattering cross sections represented by $f(\eta) = (\eta^2/\pi a^2) \times (d\sigma/d\eta)$ as functions of $\eta = \varepsilon \sin(\theta/2)$. The solid curves were computed exactly from classical mechanics for different values of ε . The curve designated by "M" represents the cross section used by Molière [4]. The Rutherford cross section is designated by "R". The dashed line on the left is approached by $f(\eta)$ for low values of η .

the energies $\varepsilon = 10^{-3}, 10^{-2}, 10^{-1}$, and 1 the cross sections calculated by Robinson [2] were used, and for $\varepsilon = 10$ computations by the present author. As can be seen in Fig. 1, the exactly computed functions deviate somewhat from the universal function $f(\eta)$ in the vicinity of $\eta = \varepsilon$ (i.e. $\theta = \pi$). For small angles, however, they asymptotically approach $f(\eta)$. At $\theta = 0.1$ (i.e. $\eta = 0.05 \varepsilon$) the deviation is generally smaller than 1%.

Molière [4], in his well-known theory of multiple scattering, used the scattering cross section

$$\frac{d\sigma}{d\theta} = 2\pi b^2 \frac{\theta}{(\theta_a^2 + \theta^2)^2}, \quad (5)$$

where

$$\theta_a^2 = \frac{\lambda_{\text{Br}}^2}{a^2} (1.13 + 3.76 \alpha^2) \quad (6)$$

and $\lambda_{\text{Br}} = \hbar(m_1 + m_2)/(m_1 m_2 v)$ is the de Broglie wavelength.

With $\theta_a^2 = \lambda_{\text{Br}}^2/a^2$, equation (5) corresponds to the first Born approximation for an exponentially screened potential. This would be valid for $\alpha \ll 1$. With the assumption (6) Molière attempted to extend the validity of (5) to larger values of α . This extension, however, does not carry very far. For $\alpha^2 \gg 1.13/3.76 = 0.3$ the scattering cross section (5), (6) in the form of equation (4) gives a function

$$f_M(\eta) = \frac{\eta^3}{2 \left(\frac{3.76}{16} + \eta^2 \right)^2}. \quad (5a)$$

This function is shown in Fig. 1, designated by "M". For $\eta < 0.5$ there are very strong deviations from the exact cross sections. The first Born approximation gives similar functions $f(\eta)$, which have their maximum value, however, in the vicinity of $\eta = (2\alpha)^{-1}$.

The strong decline of these scattering cross sections with decreasing angles results in finite total scattering cross sections. The differential cross section (5), (6), used by Molière, e.g. results (for $\alpha^2 \gg 0.3$) in a total cross section

$$\sigma_M = 1.06 \pi a^2 \frac{\pi \varepsilon^2}{0.94 + \pi^2 \varepsilon^2} < 1.06 \pi a^2 < 5 \times 10^{-17} \text{ cm}^2.$$

This cross section is therefore always small compared with the area occupied by an atom in a monoatomic layer of a solid body (10^{-15} cm^2). Consequently the number of collisions, encountered by a particle in Molière's theory, is very much smaller than the number of atomic layers penetrated.

The differential scattering cross section (4) calculated from classical mechanics, however, results in a diverging total cross section which has to be cut off on physical grounds.

In the following computation the scattering cross section (4) is used. The function $f(\eta)$ is given in Table 1. The numerical values of this table were calculated by the author, using in part the tables of Robinson [2]. They are in good accord with the values given by Lindhard, Nielsen, and Scharff [3].

3. Calculation of the Angular Distribution of the Scattered Particles

We consider a beam of initially parallel particles with mass m_1 and atomic number Z_1 , which penetrates a material layer of thickness t with N atoms per unit volume of mass m_2 and atomic number Z_2 . We assume that each scattering

centre will be effective according to the scattering cross section (4) within a spherical volume of radius r_0 .

In correspondence to the properties of a solid body we put r_0 equal to half the distance of immediately neighbouring atoms,

$$r_0 \approx \frac{1}{2} N^{-1/3} \quad (7)$$

The restriction to this volume results in a minimum scattering angle η_0 for single scattering, which is determined by

$$\int_{\eta_0}^{\epsilon} \frac{d\sigma}{d\eta} d\eta = \pi r_0^2. \quad (8)$$

The following calculation will show that the results do not depend in a critical manner on the exact values of r_0 and η_0 . Taking as a basis the conditions in an amorphous or very fine polycrystalline solid body, we assume the impact parameters to be uniformly distributed over the cross section πr_0^2 . Thus we obtain for the average number of collisions, n , suffered by a particle penetrating a layer of thickness t ,

$$n = \pi r_0^2 N t. \quad (9)$$

As the spherical volumes of radius r_0 fill the volume of the solid layer very tightly, the number of collisions cannot fluctuate very much about the value (9). Therefore, at first we shall consider the number of collisions (9) to be constant, taking into account the influence of possible fluctuations later on.

After the first collision the fraction of particles, being scattered into the solid angle $d\omega$ around the scattering angle ϑ in the laboratory system, is given by

$$F_1(\vartheta) d\omega = \begin{cases} \frac{1}{\pi r_0^2} \frac{d\sigma}{d\omega} d\omega & \text{for } \eta \geq \eta_0, \\ 0 & \text{for } \eta < \eta_0. \end{cases} \quad (10)$$

Using equation (4) and the relation $\vartheta = \theta m_2/(m_1 + m_2) = 2\eta m_2/\epsilon(m_1 + m_2)$, holding for small angles θ , we get for the scattering cross section $d\sigma/d\omega$ in the laboratory system

$$\frac{d\sigma}{d\omega} = \frac{\epsilon^2}{8} a^2 \left(\frac{m_1 + m_2}{m_2} \right)^2 \frac{f(\eta)}{\eta^3}. \quad (11)$$

The further treatment follows the lines of Molière's computation [4]. After the second collision we get the distribution

$$F_2(\vartheta) d\omega = d\omega \int_0^\infty \int_0^{2\pi} F_1(\vartheta - \chi) F_1(\chi) d\omega_\chi. \quad (12)$$

In this expression the angles ϑ and χ are interpreted as vectors within a plane perpendicular to the direction of incidence within unit distance from the scattering layer. The integration extends over the elements of area $d\omega_\chi$ in this

²⁾ To be more precise we have $r_0 = 0.562 N^{-1/3}$ for an f.c.c. lattice, $r_0 = 0.545 N^{-1/3}$ for a b.c.c. lattice, $r_0 = 0.434 N^{-1/3}$ for a diamond-type lattice, and $r_0 = 0.562 N^{-1/3}$ for the hexagonal maximum density case.

plane, pertaining to the angle χ . After the n -th collision we get the final distribution

$$F_n(\boldsymbol{\theta}) d\omega = d\omega \int_0^\infty \int_0^{2\pi} F_{n-1}(\boldsymbol{\theta} - \boldsymbol{\chi}) F_1(\boldsymbol{\chi}) d\omega_\chi. \quad (13)$$

From this expression we get, in a manner given in detail by Molière [4], using a cylindric symmetrical Fourier-like transformation,

$$F_n(\boldsymbol{\theta}) d\omega = \frac{d\omega}{2\pi} \int_0^\infty G_1^n(y) J_0(\boldsymbol{\theta} y) y dy, \quad (14)$$

where

$$G_1(y) = 2\pi \int_0^\infty F_1(x) J_0(yx) x dx.$$

J_0 is the Bessel function of zero order, x and y are auxiliary variables.

Using $F_1(x)$ from (10) and (11) and transforming for the new variable $z = 2y(m_1 + m_2)/(\varepsilon m_2)$ we get

$$G_1(z) = \frac{a^2}{r_0^2} \int_{\eta_0}^\infty \frac{f(\eta)}{\eta^2} J_0(z\eta) d\eta$$

and

$$F_n(\tilde{\boldsymbol{\theta}}) d\omega = \frac{d\omega}{2\pi} \frac{\varepsilon^2}{4} \left(\frac{m_1 + m_2}{m_2} \right)^2 \int_0^\infty G_1^n(z) J_0(\tilde{\boldsymbol{\theta}} z) z dz. \quad (15)$$

In this expression, $\boldsymbol{\theta}$ has been replaced by the variable

$$\tilde{\boldsymbol{\theta}} = \boldsymbol{\theta} \frac{\varepsilon}{2} \frac{m_1 + m_2}{m_2}, \quad (16)$$

which shall be denoted as reduced scattering angle.

The function $G_1(z)$ now additionally depends on η_0 which, according to (8), is again determined by r_0 . In order to eliminate this dependence on η_0 we set up the function

$$\Delta(z) = \frac{r_0^2}{a^2} (1 - G_1(z)) = \int_{\eta_0}^\infty \frac{f(\eta)}{\eta^2} [1 - J_0(z\eta)] d\eta. \quad (17)$$

In this integral the integrand behaves like $(z^2/4)f(\eta)$ for small values of η . With η increasing, the integrand increases from zero, then goes through a maximum and again drops to zero, somewhat oscillating. For $z < 10$ the maximum of the integrand is determined by the maximum of $f(\eta)$ and is attained at $\eta_{\max} = 0.15$. As can be shown from (8), in all cases of practical importance there is always $\eta_0 < 0.03$. Therefore there will be only a slight error if in (17) we replace the lower limit of integration, η_0 , by zero. With this substitution, the function $\Delta(z)$ has been calculated from (17) by numerical integration in the interval $0.05 \leq z \leq 100$. With $G_1(z) = 1 - (a^2/r_0^2)\Delta(z)$ we get from (15)

$$F_n(\tilde{\boldsymbol{\theta}}) d\omega = \frac{d\omega}{2\pi} \frac{\varepsilon^2}{4} \left(\frac{m_1 + m_2}{m_2} \right)^2 \int_0^\infty \left[1 - \frac{a^2}{r_0^2} \Delta(z) \right]^n J_0(\tilde{\boldsymbol{\theta}} z) z dz. \quad (18)$$

In this representation of the scattering distribution the evaluation of the integral is complicated by its dependence on the two parameters a^2/r_0^2 and $n = \pi r_0^2 N t$. For reduction of this twofold dependence we put

$$\left[1 - \frac{a^2}{r_0^2} \Delta\right]^n = \left[1 - \frac{1}{2} \frac{a^2}{r_0^2} \tau \Delta^2\right] \exp(-\tau \Delta), \quad (19)$$

where

$$\tau = \frac{a^2}{r_0^2} n = \pi a^2 N t. \quad (20)$$

Equation (19) is a good approximation if $(a^2/r_0^2) \Delta(z) \ll 1$, which is always the case. The quantity τ defined by (20) shall be denoted as reduced thickness of the scattering layer.

Using (19) we get from (18)

$$F_n(\tilde{\vartheta}) d\omega = \frac{d\omega}{2\pi} \frac{\varepsilon^2}{4} \left(\frac{m_1 + m_2}{m_2}\right)^2 \left[f_1(\tau, \tilde{\vartheta}) - \frac{a^2}{r_0^2} f_2(\tau, \tilde{\vartheta})\right], \quad (21)$$

where

$$f_1(\tau, \tilde{\vartheta}) = \int_0^\infty \exp[-\tau \Delta(z)] J_0(\tilde{\vartheta} z) z dz \quad (21a)$$

and

$$f_2(\tau, \tilde{\vartheta}) = \frac{\tau}{2} \int_0^\infty \exp[-\tau \Delta(z)] J_0(\tilde{\vartheta} z) \Delta^2(z) z dz. \quad (21b)$$

The functions f_1 and f_2 are only dependent on the reduced scattering angle $\tilde{\vartheta}$ and the reduced thickness τ . Both functions have been calculated by numerical integration in the τ -interval $0.2 \leq \tau \leq 20$.

The angular distributions $F_n(\tilde{\vartheta})$ are primarily determined by the function f_1 whereas the term $(a^2/r_0^2) f_2$ generally is only a small correction. This means that the angular distribution is essentially determined by the reduced thickness τ , whereas the radius r_0 of the spheres of interaction influences only the small correction term.

The numerical results for the function f_1 are compiled in Table 2. The function f_2 is given in Table 3. For computational reasons this latter function has not been calculated from equation (21b), but from the relation

$$f_2 = \frac{n}{\tau} (f_1 - f_3),$$

where

$$f_3 = \int_0^\infty \left[1 - \frac{\tau}{n} \Delta(z)\right]^n J_0(\tilde{\vartheta} z) dz.$$

In the numerical computation the value $n = 50$ was used. The results are, however, independent of this special value.

For low thicknesses τ the angular distributions show a bump in the vicinity of $\tilde{\vartheta} = 9$. This bump is barely visible at $\tau = 12$, but more prominent for smaller thicknesses. This apparently is the same phenomenon as has already been detected by Marion and Zimmerman [6] in the angular distributions calculated according to Molière [4] as well as according to Nigam, Sundaresan, and Wu [7].

Table 2
The function $f_1(\tau, \tilde{\theta})$.
The last number and its sign designate the power of ten by which the preceding number is to be multiplied

$\tilde{\theta}$	τ								
	0.2	0.4	0.6	0.8	1.0	1.2	1.4	1.6	1.8
0	1.96+2	4.55+1	2.11+1	1.25+1	8.48+0	6.21+0	4.80+0	3.86+0	3.20+0
0.05	9.82+1	3.72+1	1.97+1	1.20+1	8.27+0	6.11+0	4.74+0	3.83+0	3.17+0
0.1	3.96+1	2.58+1	1.65+1	1.09+1	7.73+0	5.82+0	4.58+0	3.72+0	3.10+0
0.15	1.76+1	1.58+1	1.27+1	9.26+0	6.93+0	5.38+0	4.31+0	3.55+0	2.99+0
0.2	8.62+0	1.01+1	9.45+0	7.58+0	6.02+0	4.85+0	3.98+0	3.33+0	2.84+0
0.25	4.65+0	6.55+0	6.91+0	6.06+0	5.11+0	4.28+0	3.62+0	3.08+0	2.66+0
0.3	2.74+0	4.45+0	5.03+0	4.78+0	4.27+0	3.72+0	3.23+0	2.82+0	2.47+0
0.35	1.77+0	3.02+0	3.71+0	3.76+0	3.53+0	3.20+0	2.86+0	2.55+0	2.27+0
0.4	1.22+0	2.19+0	2.78+0	2.96+0	2.91+0	2.73+0	2.51+0	2.28+0	2.07+0
0.45	8.82-1	1.59+0	2.12+0	2.35+0	2.39+0	2.32+0	2.19+0	2.03+0	1.87+0
0.5	6.55-1	1.20+0	1.64+0	1.88+0	1.97+0	1.96+0	1.90+0	1.79+0	1.68+0
0.6	3.80-1	7.15-1	1.01+0	1.22+0	1.35+0	1.40+0	1.41+0	1.39+0	1.34+0
0.7	2.26-1	4.45-1	6.44-1	8.08-1	9.28-1	1.01+0	1.05+0	1.06+0	1.05+0
0.8	1.39-1	2.80-1	4.21-1	5.45-1	6.46-1	7.22-1	7.75-1	8.07-1	8.21-1
0.9	8.22-2	1.76-1	2.78-1	3.71-1	4.53-1	5.21-1	5.74-1	6.12-1	6.37-1
1.0	5.04-2	1.11-1	1.86-1	2.57-1	3.22-1	3.79-1	4.27-1	4.65-1	4.94-1
1.2	2.51-2	5.60-2	9.24-2	1.31-1	1.69-1	2.02-1	2.40-1	2.71-1	2.97-1
1.4	1.52-2	3.20-2	5.08-2	7.23-2	9.51-2	1.18-1	1.41-1	1.63-1	1.83-1
1.6	1.03-2	2.05-2	3.22-2	4.55-2	6.01-2	7.53-2	9.02-2	1.05-1	1.19-1
1.8	8.8-3	1.48-2	2.25-2	3.13-2	4.01-2	5.03-2	6.01-2	7.01-2	8.01-2
2.0	6.1-3	1.15-2	1.71-2	2.28-2	2.89-2	3.52-2	4.18-2	4.86-2	5.55-2
2.5						1.71-2	1.98-2	2.28-2	2.58-2
3.0						8.9-3	1.02-2	1.16-2	1.31-2
3.5						4.9-3	5.7-3	6.4-3	7.2-3
4.0						2.9-3	3.4-3	3.9-3	4.3-3

$\tilde{\theta}$	τ								
	2.0	2.5	3.0	3.5	4.0	4.5	5.0	6.0	7.0
0	2.71+0	1.92+0	1.46+0	1.16+0	9.52-1	8.03-1	6.90-1	5.32-1	4.28-1
0.2	2.45+0	1.79+0	1.39+0	1.12+0	9.23-1	7.82-1	6.75-1	5.23-1	4.23-1
0.4	1.87+0	1.48+0	1.20+0	9.96-1	8.42-1	7.24-1	6.32-1	4.98-1	4.07-1
0.5	1.56+0	1.30+0	1.09+0	9.19-1	7.89-1	6.86-1	6.03-1	4.80-1	3.95-1
0.6	1.28+0	1.11+0	9.62-1	8.33-1	7.27-1	6.40-1	5.69-1	4.59-1	3.81-1
0.8	8.23-1	7.90-1	7.29-1	6.64-1	6.01-1	5.44-1	4.94-1	4.12-1	3.49-1
1.0	5.14-1	5.36-1	5.29-1	5.07-1	4.78-1	4.47-1	4.16-1	3.60-1	3.13-1
1.2	3.19-1	3.58-1	3.76-1	3.78-1	3.70-1	3.57-1	3.45-1	3.08-1	2.76-1
1.4	2.02-1	2.40-1	2.64-1	2.77-1	2.82-1	2.80-1	2.65-1	2.59-1	2.39-1
1.5	1.67-1	1.96-1	2.20-1	2.36-1	2.44-1	2.47-1	2.45-1	2.35-1	2.21-1
1.6	1.33-1	1.61-1	1.85-1	2.02-1	2.12-1	2.18-1	2.18-1	2.14-1	2.03-1
1.8	8.99-2	1.12-1	1.32-1	1.48-1	1.59-1	1.67-1	1.68-1	1.75-1	1.72-1
2.0	6.24-2	7.94-2	9.50-2	1.09-1	1.20-1	1.29-1	1.35-1	1.42-1	1.43-1
2.2	4.55-2	5.74-2	6.98-2	8.11-2	9.09-2	9.92-2	1.06-1	1.15-1	1.19-1
2.4	3.85-2	4.22-2	5.19-2	6.11-2	6.95-2	7.69-2	8.33-2	9.28-2	9.85-2
2.6	2.50-2	3.16-2	3.92-2	4.66-2	5.35-2	6.00-2	6.57-2	7.49-2	8.13-2
2.8	1.90-2	2.40-2	2.99-2	3.58-2	4.16-2	4.70-2	5.20-2	6.05-2	6.70-2
3.0	1.47-2	1.86-2	2.32-2	2.79-2	3.25-2	3.70-2	4.12-2	4.89-2	5.51-2
3.5	8.1-3	1.04-2	1.30-2	1.57-2	1.84-2	2.12-1	2.40-2	2.93-2	3.42-2
4.0	4.8-3	6.2-3	7.7-3	9.3-3	1.09-2	1.26-2	1.44-2	1.79-2	2.14-2
4.5	2.8-3	3.8-3	4.7-3	5.7-3	6.7-3	7.5-3	8.9-3	1.13-2	1.36-2
5.0	1.7-3	2.3-3	2.9-3	3.6-3	4.2-3	4.9-3	5.6-3	7.2-3	8.8-3
6.0							2.0-3	2.8-3	3.5-3
7.0							8.8-4	1.2-3	1.6-3

Table 2 (continued)

$\tilde{\vartheta}$	τ						
	8.0	10	12	14	16	18	20
0	3.65-1	2.62-1	2.05-1	1.67-1	1.41-1	1.21-1	1.05-1
0.5	3.33-1	2.50-1	1.95-1	1.61-1	1.36-1	1.18-1	1.03-1
1.0	2.75-1	2.18-1	1.76-1	1.48-1	1.27-1	1.11-1	9.80-2
1.5	2.04-1	1.75-1	1.50-1	1.29-1	1.13-1	1.01-1	9.00-2
2.0	1.41-1	1.31-1	1.19-1	1.08-1	9.71-2	8.80-2	8.01-2
2.5	9.32-2	9.42-2	9.10-2	8.75-2	8.00-2	7.44-2	6.91-2
3.0	5.98-2	6.52-2	6.72-2	6.62-2	6.40-2	6.12-2	5.82-2
3.5	3.83-2	4.45-2	4.80-2	4.96-2	4.98-2	4.90-2	4.77-2
4.0	2.46-2	3.01-2	3.40-2	3.65-2	3.79-2	3.84-2	3.83-2
4.5	1.59-2	2.03-2	2.39-2	2.66-2	2.85-2	2.97-2	3.04-2
5.0	1.04-2	1.37-2	1.66-2	1.92-2	2.12-2	2.27-2	2.37-2
6.0	4.39-3	6.26-3	8.26-3	9.96-3	1.15-2	1.29-2	1.41-2
7.0	2.06-3	3.02-3	4.24-3	5.28-3	6.32-3	7.32-3	8.26-3
8.0	1.21-3	1.69-3	2.24-3	2.85-3	3.50-3	4.16-3	4.82-3
9.0	8.5-4	1.10-3	1.38-3	1.65-3	2.03-3	2.45-3	2.88-3
10	5.9-4	7.4-4	8.50-4	9.90-4	1.23-3	1.49-3	1.71-3
11	3.9-4	4.6-4	5.20-4	6.30-4	7.65-4	9.65-4	1.12-3
12	2.4-4	2.7-4	3.10-4	3.98-4	4.97-4	6.03-4	7.18-4
13	1.5-4	1.7-4	2.15-4	2.70-4	3.35-4	4.35-4	5.00-4
14	1.0-4	1.2-4	1.46-4	1.90-4	2.40-4	2.88-4	3.43-4
15			1.04-4	1.41-4	1.80-4	2.10-4	2.50-4
16			8.2-5	1.06-4	1.38-4	1.58-4	1.85-4
18			5.4-5	7.0-5	8.6-5	1.03-4	1.20-4
20			4.2-5	5.4-5	6.5-5	7.7-5	8.8-5

Table 3

The function $f_2(\tau, \tilde{\vartheta})$.

The last number and its sign designate the power of ten by which the preceding number is to be multiplied

$\tilde{\vartheta}$	τ								
	0.2	0.4	0.6	0.8	1.0	1.2	1.4	1.6	1.8
0	+3.52+3	+3.27+2	+9.08+1	+3.85+1	+2.00+1	+1.18+1	+7.55+0	+5.16+0	+3.71+0
0.05	+2.58+2	+1.63+2	+7.30+1	+3.42+1	+1.85+1	+1.11+1	+7.18+0	+4.96+0	+3.59+0
0.1	-1.12+2	+4.84+0	+3.56+1	+2.34+1	+1.45+1	+9.33+0	+6.37+0	+4.51+0	+3.32+0
0.15	-5.60+1	-1.12+1	+9.87+0	+1.24+1	+9.59+0	+7.01+0	+5.16+0	+3.83+0	+2.91+0
0.2	-2.13+1	-1.22+1	-2.23+0	+3.88+0	+5.15+0	+4.65+0	+3.87+0	+3.12+0	+2.45+0
0.25	-8.25+0	-9.58+0	-5.59+0	-1.40+0	+1.76+0	+2.71+0	+2.71+0	+2.35+0	+1.95+0
0.3	-3.22+0	-6.12+0	-5.28+0	-2.87+0	-1.92-1	+1.32+0	+1.69+0	+1.74+0	+1.48+0
0.35	-1.11+0	-3.40+0	-4.12+0	-3.08+0	-6.30-1	+3.60-1	+9.20-1	+1.03+0	+1.04+0
0.4	-2.27-1	-2.00+0	-2.93+0	-2.69+0	-1.48+0	-3.14-1	+2.69-1	+5.28-1	+6.09-1
0.45	+1.54-1	-1.09+0	-2.10+0	-2.15+0	-1.47+0	-6.77-1	-1.80-1	+1.08-1	+2.70-1
0.5	+3.28-1	-6.30-1	-1.50+0	-1.68+0	-1.34+0	-8.43-1	-4.60-1	-1.85-1	-4.67-3
0.6	+3.32-1	-2.06-1	-7.32-1	-9.90-1	-9.42-1	-8.20-1	-6.06-1	-4.51-1	-3.01-1
0.7	+2.72-1	-3.34-2	-3.49-1	-5.65-1	-6.03-1	-5.79-1	-5.05-1	-4.31-1	-3.45-1
0.8	+2.02-1	+2.80-2	-1.54-1	-3.00-1	-3.59-1	-3.76-1	-4.60-1	-3.40-1	-3.08-1
0.9	+1.38-1	+4.84-2	-5.56-2	-1.44-1	-2.04-1	-2.39-1	-2.54-1	-2.49-1	-2.48-1
1.0	+9.47-2	+4.86-2	-1.08-2	-6.44-2	-1.02-1	-1.34-1	-1.62-1	-1.79-1	-1.87-1
1.2	+5.33-2	+3.71-2	+1.85-2	+1.63-3	-1.69-2	-3.69-2	-5.66-2	-7.78-2	-9.33-2
1.4	+3.38-2	+2.40-2	+1.62-2	+9.9-3	+3.76-3	-4.93-3	-1.66-2	-3.05-2	-4.22-2
1.6	+2.12-2	+1.56-2	+1.05-2	+7.8-3	+7.92-3	+6.3-3	+3.2-4	-8.5-3	-1.66-2
1.8	+1.40-2	+9.2-3	+5.3-3	+4.7-3	+6.81-3	+8.4-3	+5.3-3	+8.8-4	-3.3-3
2.0	+9.2-3	+4.7-3	+1.7-3	+2.6-3	+4.49-3	+6.6-3	+6.0-3	+4.7-3	+2.8-3

Table 3 (continued)

$\tilde{\phi}$	τ								
	2.0	2.5	3.0	3.5	4.0	4.5	5.0	6.0	7.0
0	+2.75+0	+1.94+0	+9.13-1	+6.06-1	+4.26-1	+3.14-1	+2.40-1	+1.51-1	+1.03-1
0.2	+1.94+0	+1.16+0	+7.56-1	+5.26-1	+3.81-1	+2.87-1	+2.23-1	+1.43-1	+9.78-2
0.4	+5.85-1	+5.04-1	+4.10-1	+3.30-1	+2.69-1	+2.17-1	+1.78-1	+1.22-1	+8.71-2
0.5	+7.83-2	+2.00-1	+2.35-1	+2.19-1	+1.97-1	+1.73-1	+1.48-1	+1.08-1	+7.93-2
0.6	-1.82-1	+1.56-2	+1.04-1	+1.36-1	+1.38-1	+1.31-1	+1.19-1	+9.46-2	+7.19-2
0.8	-2.71-1	-1.66-1	-7.29-2	-4.74-3	+3.60-2	+5.50-2	+6.28-2	+5.98-2	+5.09-2
1.0	-1.87-1	-1.58-1	-1.09-1	-5.80-2	-2.03-2	+2.48-3	+1.99-2	+3.36-2	+3.27-2
1.2	-1.01-1	-1.05-1	-8.95-2	-6.63-2	-3.93-2	-2.38-2	-9.22-3	+8.47+3	+1.52-2
1.4	-5.19-2	-6.47-2	-6.51-2	-5.62-2	-4.51-2	-3.49-2	-2.45-2	-8.19-3	+2.05-3
1.5	-3.68-2	-4.89-2	-5.36-2	-5.06-2	-4.27-2	-3.65-2	-2.80-2	-1.33-2	-3.47-3
1.6	-2.33-2	-3.69-2	-4.41-2	-4.38-2	-3.97-2	-3.50-2	-2.88-2	-1.60-2	-6.68-3
1.8	-8.76-3	-2.07-2	-2.90-2	-3.17-2	-3.09-2	-2.92-2	-2.63-2	-1.79-2	-1.03-2
2.0	-1.20-3	-1.11-2	-1.90-2	-2.20-2	-2.32-2	-2.24-2	-2.10-2	-1.66-2	-1.11-2
2.2	+1.72-3	-4.82-3	-1.02-2	-1.42-2	-1.65-2	-1.66-2	-1.60-2	-1.39-2	-1.09-2
2.4	+2.68-3	-1.18-3	-5.19-3	-8.30-5	-1.01-2	-1.14-2	-1.16-2	-1.16-2	-9.99-3
2.6	+2.81-3	+8.21-4	-1.96-3	-3.99-3	-5.89-3	-7.13-3	-8.15-3	-9.05-3	-8.60-3
2.8	+2.61-3	+1.35-3	-2.99-4	-1.79-3	-3.12-3	-4.44-3	-5.61-3	-7.01-3	-7.27-3
3.0	+2.06-3	+1.45-3	+4.64-4	-5.97-4	-1.71-3	-2.79-3	-3.84-3	-5.29-3	-5.90-3
3.5	+1.07-3	+9.39-4	+8.22-4	+3.58-4	-1.15-4	-6.60-4	-1.18-3	-2.15-3	-2.88-3
4.0	+4.97-4	+5.46-4	+6.15-4	+5.56-4	+3.14-4	+9.80-5	-1.30-4	-5.98-4	-1.07-4
4.5	+1.85-4	+3.11-4	+4.25-4	+4.08-4	+3.63-4	+3.04-4	+2.24-4	+2.8-5	-2.1-4
5.0	+4.8-5	+1.48-4	+2.44-4	+2.80-4	+3.01-4	+3.11-4	+3.13-4	+2.4-4	+1.1-4
6.0							+1.39-4	+1.8-4	+1.8-4
7.0							+4.38-5	+7.3-5	+8.4-5

$\tilde{\phi}$	τ						
	8	10	12	14	16	18	20
0	+7.36-2	+4.21-2	+2.69-2	+1.83-2	+1.34-2	+1.01-2	+7.88-3
0.5	+5.79-2	+3.61-2	+2.34-2	+1.64-2	+1.21-2	+9.26-3	+7.28-3
1.0	+2.94-2	+2.17-2	+1.60-2	+1.23-2	+9.49-3	+7.45-3	+5.95-3
1.5	+2.30-3	+7.07-3	+7.76-3	+7.02-3	+6.13-3	+5.17-3	+4.34-3
2.0	-7.50-3	-2.00-3	+9.93-4	+2.36-3	+2.82-3	+2.86-3	+2.72-3
2.5	-8.27-3	-5.37-3	-2.58-3	-7.96-4	+3.75-4	+9.71-4	+1.28-3
3.0	-5.79-3	-5.12-3	-3.86-3	-2.46-3	-1.20-3	-3.74-4	+1.74-4
3.5	-3.26-3	-3.43-3	-3.26-3	-2.68-3	-1.84-3	-1.12-3	-4.54-4
4.0	-1.46-3	-1.49-3	-2.20-3	-2.18-3	-1.85-3	-1.40-3	-8.15-4
4.5	-4.29-4	-9.44-4	-1.29-3	-1.50-3	-1.51-3	-1.36-3	-9.57-4
5.0	-3.3-5	-3.66-4	-6.78-4	-9.38-4	-1.09-3	-1.09-3	-9.56-4
6.0	+1.5-4	+3.1-5	-1.38-4	-3.06-4	-4.67-4	-5.48-4	-6.08-4
7.0	+1.0-4	+8.5-5	+2.3-5	-6.6-5	-1.58-4	-2.40-4	-3.05-4
8.0	+5.4-5	+6.5-5	+4.9-5	+1.2-5	-3.6-5	-8.9-5	-1.31-4
9.0	+2.9-5	+4.3-5	+4.4-5	+2.9-5	+5.1-6	-2.2-5	-4.8-5
10	+1.4-5	+2.4-5	+2.8-5	+2.6-5	+1.9-5	+7.5-6	-1.1-5

Very prominent bumps only occur at very low values of the function f_1 , which have not been included in Table 2.

In the derivation of equation (18) there were no essential restrictions as to the number n of collisions. A more detailed examination of the replacement of η_0 by zero in (17), however, shows that this replacement is tied to the condition $n \gg 20 (a/r_0)^{4/3}$, which may be of some importance for the very largest values

of a/r_0 . Such values occur when carbon foils are bombarded by very light particles, e.g. protons ($a/r_0 = 0.26$). Apart from this extreme case, equation (18) should be applicable to plural scattering ($n < 20$) as well as to multiple scattering ($n > 20$). A more detailed examination of the approximation (19) further reveals that this equation, and henceforth equation (21), is tied to the condition $n \gg 5$. A comparison at $\tau = 0.2$ and $\tau = 1$ revealed deviations of the values computed according to (21) from those calculated according to (18) of less than 8% for $n = 10$, and of less than 2% for $n = 20$.

4. Discussion

First of all some general conclusions can be drawn from equation (21).

The form of the angular distribution is independent of the energy of the incident particles and primarily determined by the reduced scattering thickness τ .

The width of the distributions, measured by the reduced scattering angle $\tilde{\vartheta}$, according to (21) is also independent of the energy. Some minor dependence on energy may, however, be caused by a variation of the mean charge carried by the incident particles, which changes the effective screening parameter a and hence also $\tau = \pi a^2 N t$. As this variation of charge will only affect the incident particle, and only its outermost electrons, the influence on the parameter a should remain small. The width of the distribution, measured by the angle $\tilde{\vartheta}_{1/2}$, at which the intensity has dropped to half its value at $\tilde{\vartheta} = 0$, therefore at most might show some slight increase with increasing energy. The half-width $\tilde{\vartheta}_{1/2} = \tilde{\vartheta}_{1/2}(2/\varepsilon) m_2/(m_1 + m_2)$, as directly measured by experiment, consequently is expected to be inversely proportional to the energy.

Fig. 2 shows the half-width $\tilde{\vartheta}_{1/2}$ as a function of the reduced thickness τ . The solid curves refer to the case $a/r_0 = 0$ (lower curve) and to the case $a/r_0 = 0.26$ (upper curve), which represents the largest value of a/r_0 occurring in practice. As can be seen, the parameter a/r_0 generally is of slight influence on the half-width; only at very small thicknesses τ and large values of a/r_0 there are important deviations from the curve for $a/r_0 = 0$.

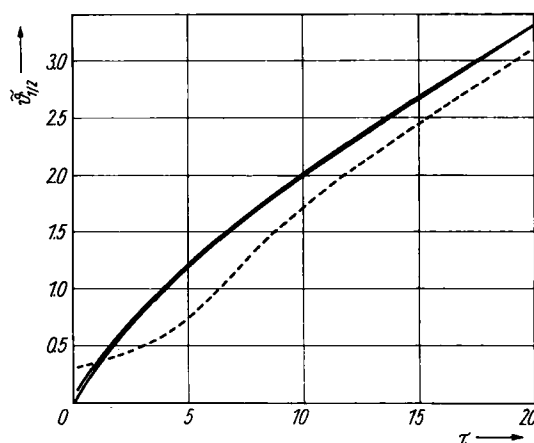


Fig. 2. Reduced half-width $\tilde{\vartheta}_{1/2}$ as function of the reduced thickness τ . The solid curves refer to the cases $a/r_0 = 0$ (lower curve) and $a/r_0 = 0.26$ (upper curve). The dashed curve represents the half-width according to Keil, Zettler, and Zinn [8].

Table 4
The functions $g_1(\tau)$ and $g_2(\tau)$

τ	0.2	0.4	0.6	0.8	1.0	1.2	1.4	1.6	1.8
$g_1(\tau)$	0.050	0.115	0.183	0.245	0.305	0.363	0.419	0.473	0.525
$g_2(\tau)$		1.25	0.91	0.79	0.73	0.69	0.65	0.63	0.61
τ	2.0	2.5	3.0	3.5	4.0	4.5	5.0	6.0	7.0
$g_1(\tau)$	0.575	0.689	0.799	0.905	1.01	1.10	1.19	1.37	1.54
$g_2(\tau)$	0.59	0.56	0.53	0.50	0.47	0.45	0.43	0.40	0.37
τ	8.0	9.0	10	12	14	16	18	20	
$g_1(\tau)$	1.70	1.85	1.99	2.27	2.54	2.80	3.05	3.29	
$g_2(\tau)$	0.34	0.32	0.30	0.26	0.22	0.18	0.15	0.13	

Corresponding to the form of equation (21) the half-width may be represented by

$$\tilde{\vartheta}_{1/2} = g_1(\tau) + \frac{a^2}{r_0^2} g_2(\tau) , \tag{22}$$

where the second term again is a generally small correction. The functions g_1 and g_2 are given by Table 4.

The width of the angular distribution may be influenced to some extent by fluctuations of the number of collisions, which may be caused mainly by fluctuations in the thickness of the scattering material. One obtains a very simple result if the probability for the occurrence of n collisions is given by the Poisson distribution

$$W(n) = \frac{(\bar{n})^n}{n!} \exp(-\bar{n}) , \tag{23}$$

where $\bar{n} = \pi r_0^2 N \bar{t}$ is the mean number of collisions. Using for each value of n the angular distribution (18), the resulting distribution is obtained as

$$F(\tilde{\vartheta}) \, d\omega = d\omega \sum_{n=0}^{\infty} W(n) F_n(\tilde{\vartheta}) = \frac{d\omega}{2\pi} \frac{\varepsilon^2}{4} \left(\frac{m_1 + m_2}{m_2} \right)^2 f_1(\bar{\tau}, \tilde{\vartheta}) , \tag{24}$$

where $\bar{\tau} = (a^2/r_0^2) \bar{n}$ is the reduced mean thickness and f_1 is the function defined by (21 a) and given by Table 2. The distribution (23) corresponds to the variance $(n - \bar{n})^2 = \bar{n}$. Representing different distributions with different variances as a sum or a difference of a fraction, distributed according to (23), and a fraction with the constant number of collisions $n = \bar{n}$, we get for the angular distribution of the scattered particles

$$F(\tilde{\vartheta}) \, d\omega = \frac{d\omega}{2\pi} \frac{\varepsilon^2}{4} \left(\frac{m_1 + m_2}{m_2} \right)^2 \left[f_1(\bar{\tau}, \tilde{\vartheta}) - \left(\frac{a^2}{r_0^2} - \gamma \right) f_2(\bar{\tau}, \tilde{\vartheta}) \right] , \tag{25}$$

where

$$\gamma = \frac{a^2 \overline{(n - \bar{n})^2}}{\tau_0^2 \bar{n}}.$$

If the fluctuation of the number of collisions is solely caused by fluctuations of the thickness t of the scattering layer, we get

$$\gamma = \frac{\overline{(\tau - \bar{\tau})^2}}{\bar{\tau}} = \pi a^2 N \frac{\overline{(t - \bar{t})^2}}{\bar{t}}. \quad (26)$$

Corresponding to the form of equation (25), we obtain for the half-width of the scattering distribution

$$\tilde{\vartheta}_{1/2} = g_1(\bar{\tau}) + \left(\frac{a^2}{\tau_0^2} - \gamma \right) g_2(\bar{\tau}), \quad (27)$$

where g_1 and g_2 are the functions given in Table 4.

Besides the form and the width of the angular distribution, equation (21) also determines the absolute value of the scattered intensity. $F_n(\tilde{\vartheta}) d\omega$ gives the fraction of the incident particles, which is scattered within the solid angle $d\omega$. The experimental determination of this fraction gives an additional possibility to check the theory, and particularly to determine experimentally the screening parameter a , which is given only in rough approximation by equation (3) or similar equations. A second possibility to confirm this determination of a is given by a comparison of the half-width of equation (22) with the experimental value.

5. Comparison with Other Theories

For the case of small energies, considered here, and the case of small scattering thicknesses, connected with such energies, there is available for comparison the theory of Molière [4, 5] and its further evolution for small numbers of collisions by Keil, Zeitler, and Zinn [8]. Starting from the scattering cross section (5), (6), Molière gets the scattering distribution

$$f(\vartheta_M) \vartheta_M d\vartheta_M = \vartheta_M d\vartheta_M \left[2 \exp(-\vartheta_M^2) + \frac{1}{B} f^{(1)}(\vartheta_M) + \frac{1}{B^2} f^{(2)}(\vartheta_M) + \dots \right], \quad (28)$$

where

$$\vartheta_M = \frac{\tilde{\vartheta} (m_1 + m_2)}{m_2 (B \pi b^2 N t)^{1/2}}.$$

For the case $\alpha^2 \gg 0.3$, the quantity B is determined by the equation $\exp(B)/B = 0.914 \tau$. The functions $f^{(1)}(\vartheta_M)$ and $f^{(2)}(\vartheta_M)$ have been calculated by Molière and are given in tables. The scattering angle ϑ_M and the reduced scattering angle $\tilde{\vartheta}$ used in this paper are related by $\vartheta_M = 2 \tilde{\vartheta} / (B \tau)^{1/2}$.

Equation (28) is an expansion in $1/B$ which, as Molière states, is valid for $B \geq 4.5$, which means in the present case $\tau \geq 21$. A comparison at $\tau = 20$ ($B = 4.38$) of the scattering distributions, calculated according to Molière and according to the present computation, reveals an astonishingly good agreement of the form of the distributions as well as of the absolute values of the scattered intensities. According to the relations stated above, the function $f_1(\tau, \tilde{\vartheta})$ of the present paper is to be compared with $4 f(\vartheta_M)/B \tau$. The difference of

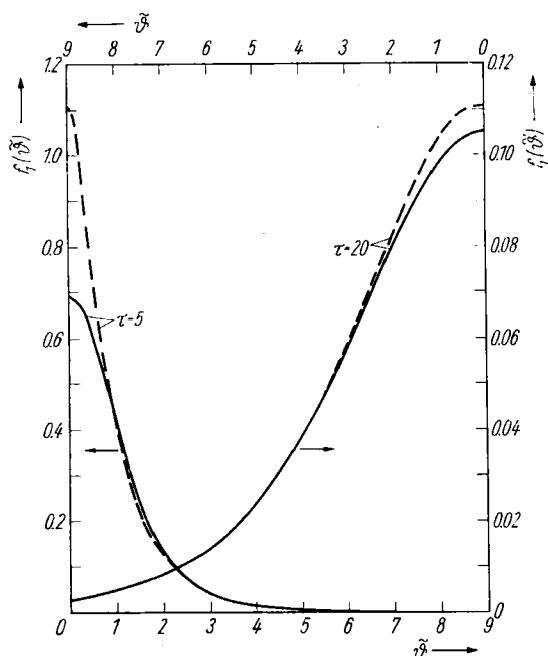


Fig. 3. Angular distributions for $\tau = 5$ and $\tau = 20$ calculated from equation (21) (solid curves) and according to Keil, Zeitler, and Zinn [8] (dashed curves)

the absolute values at $\tau = 20$ is only 0.5% for $\tilde{\vartheta} = 0$, and about 1.5% for $\tilde{\vartheta} = \tilde{\vartheta}_{1/2}$. It increases only at very large values of $\tilde{\vartheta}$. In Fig. 3, which shows the function $f_1(\tilde{\vartheta})$ for $\tau = 20$, the Molière distribution would be nearly indiscernible from this curve. At first sight this good agreement is somewhat surprising, as one would expect very much narrower distributions on the basis of the cross section (5), (6) used by Molière. The main cause of this small deviation at $\tau = 20$ seems to

be the phenomenon discussed by Bethe [9] that the distributions of multiple scattering for large numbers of collisions do not depend substantially on the exact form of the scattering cross section. The differences therefore should appear more clearly at small scattering thicknesses τ . In this domain the Molière expansion (28) is not applicable. There are, however, the calculations of Keil, Zeitler, and Zinn [8] which are likewise based on cross section (5) and which represent a direct numerical evaluation of the Molière theory for low numbers of collisions. Fig. 3 shows the scattering distributions for $\tau = 5$ and $\tau = 20$ calculated from equation (21) (solid curves) and according to Keil, Zeitler, and Zinn [8] (dashed curves). The differences for $\tau = 20$ remain comparatively small, whereas for $\tau = 5$ there are profound deviations. As expected, the scattering cross section (5), (6) results in very much narrower distributions in this range of thicknesses. This result is equally apparent in the half-widths of the scattering distributions, which are shown in Fig. 2 (dashed line). The deviations from the half-width computed in the present paper are most profound in the vicinity of $\tau = 4$. For smaller thicknesses the difference again decreases. This is caused, however, only by the fact that in the calculation of Keil, Zeitler, and Zinn for $\tau < 4$ a substantial fraction of particles penetrates the material layer without being scattered at all, whereas the half-width only relates to the scattered particles. In the vicinity of $\tau = 20$ there is a deviation, remaining nearly constant. This difference seems to be due to the approximations used by the authors cited above, which are valid only for small values of τ .

In consideration of the deviations of the scattering distributions of Keil, Zeitler, and Zinn from experimental values, these authors suggested to drop equation (6), determining the screening angle θ_a , and to adjust this quantity to the measured intensity at $\vartheta = 0$. This procedure, however, cannot be

considered satisfactory, because (in the case $\alpha > 1$ considered here) it will result in a value of θ_a which depends on the thickness of the scattering layer.

Apart from the theories considered above, the calculation of Nigam, Sundaresan, and Wu [7], at least formally, also permits an evaluation at small thicknesses and for large values of α . This theory yields scattering distributions of much larger widths than the present calculation or the theory of Molière (for large values of α). As Fleischmann [10] already has proved theoretically, and Simon [11] as well as Kerr and co-workers [12] have shown by experiment, that the theory [7] does not yield acceptable results for large values of α , no detailed comparison shall be made here.

6. Comparison with Experimental Results

The proper domain of application of the theory, derived in the present paper for the case $\alpha > 1$, is in the range of small scattering thicknesses ($\tau < 20$), which is essential for low-energy heavy particles. For larger thicknesses the distributions calculated here are nearly identical with those of the theory of Molière [4]. Therefore we restrict the comparison with experiments to the case $\tau < 20$.

Ormrod and Duckworth [13] measured the multiple scattering of neon ions with an initial energy of 61 keV ($\alpha = 8.7$) after penetration of a carbon foil of $2.12 \mu\text{g}/\text{cm}^2$. The experimental scattering distribution is shown in Fig. 4 (solid line) together with the theoretical distribution (dashed line), calculated from (21) using the approximate screening parameter $a = 1.66 \times 10^{-9}$ cm from equation (3) resulting in a reduced thickness $\tau = 0.918$, and using the mean energy $\bar{E} = 59.7$ keV (energy loss $\Delta E = 2.6$ keV) which results in a reduced energy of $\varepsilon = 4.30$. With $r_0 = 1.1 \times 10^{-8}$ cm and $a^2/r_0^2 = 0.023$ the correction term in equation (21) amounts to only a few per cent. The theoretical distribution, calculated with the approximate value of a , is distinctly narrower than the experimental one. Increasing a by 25% to the value $a = 2.09 \times 10^{-9}$ cm gives a complete agreement of the theoretical and the experimental curve within the line-width of the drawing. Unfortunately Ormrod and Duckworth do not state absolute values of the scattered intensities, so that no further check of this adjustment is possible. The calculation of Keil, Zeitler, and Zinn [8] would, in the present case, give a very strong peak in the vicinity of $\vartheta = 0$ (about 30% of the incident particles would not be scattered at all). No such peak can be seen in the experimental curve.

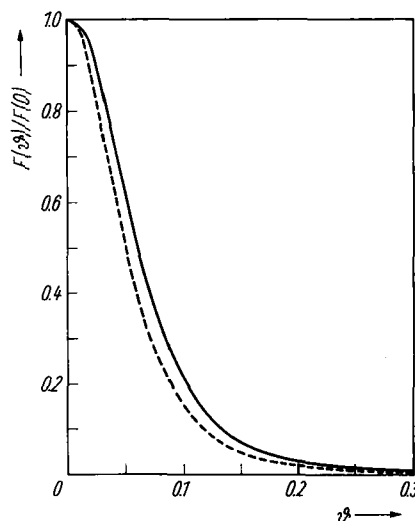


Fig. 4. Angular distribution of 61 keV neon ions after penetration of a carbon foil of $2.12 \mu\text{g}/\text{cm}^2$. The solid curve represents the measurements of Ormrod and Duckworth [13]. The dashed curve was calculated from equation (21) using the screening parameter $a = 1.66 \times 10^{-9}$ cm

Bednyakov and co-workers [14] have investigated the multiple scattering of protons with an energy of $E_0 = 163$ keV ($\alpha = 11.4$) in copper foils of different thicknesses. The measurements with foils of 190 and 248 $\mu\text{g}/\text{cm}^2$ are within the range of thicknesses to be discussed here. Using the approximate screening parameter $a = 1.54 \times 10^{-9}$ cm² calculated from equation (3) with Z_1 omitted, which is appropriate to the proton case, the reduced thicknesses are $\tau = 13.1$ and $\tau = 17.2$. Using the mean energies $\bar{E} = 1.42$ keV ($\varepsilon = 51.0$) and $\bar{E} = 134$ keV ($\varepsilon = 48.1$) respectively, the theoretical distributions calculated from equation (21) coincide with the experimental curves well within the limits of experimental error, which are stated by the authors in another paper [15] to be 2% at $\vartheta = 0$ and 3.5% at the half-width.

Bernhard and co-workers [16] have studied the multiple scattering of lithium ions in carbon and aluminium at energies between 10 and 100 keV. The thicknesses of the scattering layers were in the interval $0.7 < \tau < 10$. These authors have compared their experimental results with the calculations of the present paper, which were at their disposal prior to publication. They found good agreement between their extensive measurements and the present theory, using the screening parameters from equation (3) without any correction.

Acknowledgement

In conclusion, the author would like to express his gratitude to Prof. F. Bernhard for the constant support given to this work.

References

- [1] W. T. SCOTT, *Rev. mod. Phys.* **35**, 231 (1963).
- [2] M. T. ROBINSON, ORNL-3493, U.S. Atomic Energy Commission, 1963.
- [3] J. LINDHARD, V. NIELSEN, and M. SCHARFF, *Mat.-Fys. Medd. Dan. Vid. Selsk.* **38**, No. 10 (1968).
- [4] G. MOLIERE, *Z. Naturf.* **3a**, 78 (1948).
- [5] G. MOLIERE, *Z. Naturf.* **2a**, 133 (1947).
- [6] J. B. MARION and B. A. ZIMMERMAN, *Nuclear Instrum. and Methods* **51**, 93 (1967).
- [7] N. P. NIGAM, M. K. SUNDARESAN, and T. Y. WU, *Phys. Rev.* **115**, 491 (1959).
- [8] E. KEIL, F. ZEITLER, and W. ZINN, *Z. Naturf.* **15a**, 1031 (1960).
- [9] H. A. BETHE, *Phys. Rev.* **89**, 1256 (1953).
- [10] H. FLEISCHMANN, *Z. Naturf.* **15a**, 1090 (1960).
- [11] W. G. SIMON, *Phys. Rev.* **136**, B410 (1964).
- [12] D. KERR, G. SIEGERT, K. KÜRZINGER, E. KONECNY, and H. EWALD, *Z. Naturf.* **22a**, 1799 (1967).
- [13] J. H. ORMROD and H. E. DUCKWORTH, *Canad. J. Phys.* **41**, 1424 (1963).
- [14] A. BEDNYAKOV, V. DVOREZKII, I. SAVENKO, and A. TULINOV, *Zh. eksper. teor. Fiz.* **46**, 1901 (1964).
- [15] A. BEDNYAKOV, A. BORYAKINA, I. SAVENKO, and A. TULINOV, *Zh. eksper. teor. Fiz.* **42**, 740 (1962).
- [16] F. BERNHARD, F. LIPPOLD, L. MEYER, S. SCHWABE, and R. STOLLE, *Proc. Internat. Conf. Atomic Collision Phenomena in Solids*, Brighton, England, 1969, North-Holland Publ. Co., 1970 (p. 663).

(Received December 18, 1970)